

Review Exercise 1

Solution:

a) In F.A., $V_{BE} = 0.7 \text{ V}$. Here, $V_E = 0$, so, $V_B = V_{BE} = 0.7 \text{ V}$.

$$I_B = \frac{10 - V_B}{100k} = \frac{10 - 0.7}{100k} = 0.093 \text{ mA}$$

b) $I_C = \beta \times I_B = 50 \times 0.093 = 4.65 \text{ mA}$

c) $V_C = 20 - I_C \times 10 = 20 - 4.65 \times 10 = -26.5 \text{ V}$

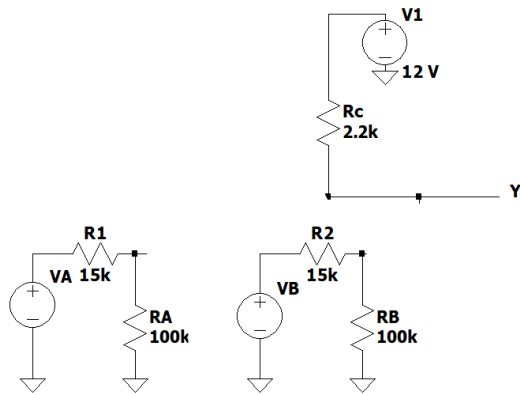
d) $V_{CE} = V_C - V_E = -26.5 - 0 = -26.5 < 0$

Exercise 1

Solution:

a) We assume cutoff mode for input logic '0' & saturation mode for input logic '1'.

Case (0,0):



Due to the open circuit, we see, $V_Y = 12 \text{ V}$
Verification:

$$V_{EQ_1} = V_{EQ_2} = 0 \text{ V}$$

$$V_A = V_B = 0.2 \text{ V}$$

Applying KCL at the base of Q_1 ,

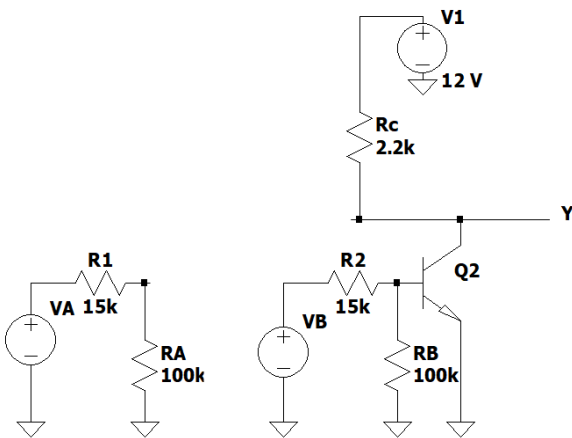
$$\frac{V_A - V_{BQ_1}}{15} = \frac{V_{BQ_1} - 0}{100k} \rightarrow V_{BQ_1} = 0.17 \text{ V}$$

$$\therefore V_{BEQ_1} = 0.17 - 0 = 0.17 \text{ V} < 0.5 \text{ V}$$

Hence Q_1 will be off

Similarly, we can show Q_2 will be off too, as it has the same parameters.

Case (0,1):



Since Q_2 will be on, we can assume it will be in saturation.

$$\therefore V_{CE_{Q_2}} = 0.2 \text{ V}$$

$$\therefore V_Y = V_{CE_{Q_2}} = 0.2 \text{ V}$$

Verification:

$$I_{C_{Q_2}} = \frac{12 - V_Y}{2.2} = 5.363 \text{ mA}$$

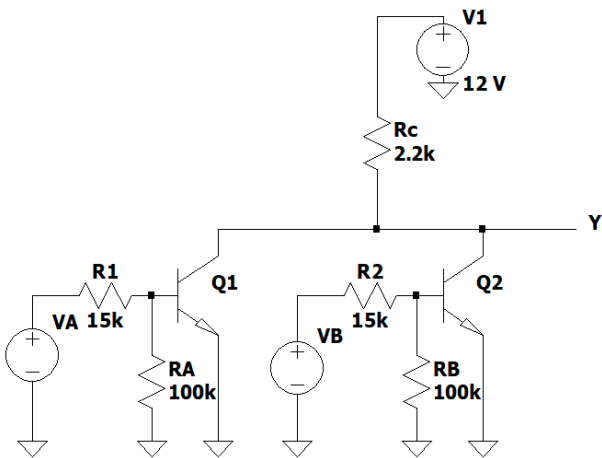
$$I_{B_{Q_2}} = I_{R_2} - I_{R_B} = \frac{12 - V_{B_{Q_2}}}{15} - \frac{V_{B_{Q_2}} - 0}{100}$$

$$= \frac{12 - 0.8}{15} - \frac{0.8}{100} = 0.738 \text{ mA}$$

$$\therefore \beta_{Q_2} = \frac{I_{C_{Q_2}}}{I_{B_{Q_2}}} = \frac{5.363}{0.738} = 7.27 < \beta_F(30)$$

Case (1,0): Same as Case (0,1) since both BJTs' parameters are identical.

Case (1,1):



$$V_Y = V_{CE_{Q_1}} = V_{CE_{Q_2}} = 0.2 \text{ V}$$

Verification:

KCL at Y,

$$I_{R_C} = I_{C_{Q_1}} + I_{C_{Q_2}} \rightarrow \frac{12 - 0.2}{2.2} = 2 \times I_{C_{Q_1}} \rightarrow$$

$$I_{C_{Q_1}} = 2.681 \text{ mA}$$

$$I_{B_{Q_1}} = I_{B_{Q_2}} =$$

$$0.738 \text{ mA [Similarly as in Case (0,1)]}$$

$$\beta_{Q_1} = \beta_{Q_2} = \frac{I_{C_{Q_1}}}{I_{B_{Q_1}}} = 3.63 < \beta_F(30)$$

Case	V_Y
0,0	12
0,1	0.2
1,0	0.2
1,1	0.2

b) High threshold = min (12) = 12 V, Low threshold = max (0.2, 0.2, 0.2) = 0.2 V

c) Applying KCL at the base of Q_2 ,

$$I_{B_{Q_2}} = I_{R_2} - I_{R_B}$$

$$= \frac{12-0.8}{15} - \frac{0.8}{2 \times 100} = 0.742 \text{ mA}$$

$$\text{Now, } \beta_{0,1} = \frac{(I_{C_{Q_2}})_{0,1}}{0.742} = \frac{5.363}{0.742} = 7.23 < \beta_F(30) \text{ \& } \beta_{1,1} = \frac{(I_{C_{Q_2}})_{1,1}}{0.742} = \frac{2.681}{0.742} = 3.61 < \beta_F(30)$$

Thus, it will still satisfy saturation mode conditions for (1,1) & (0,1).

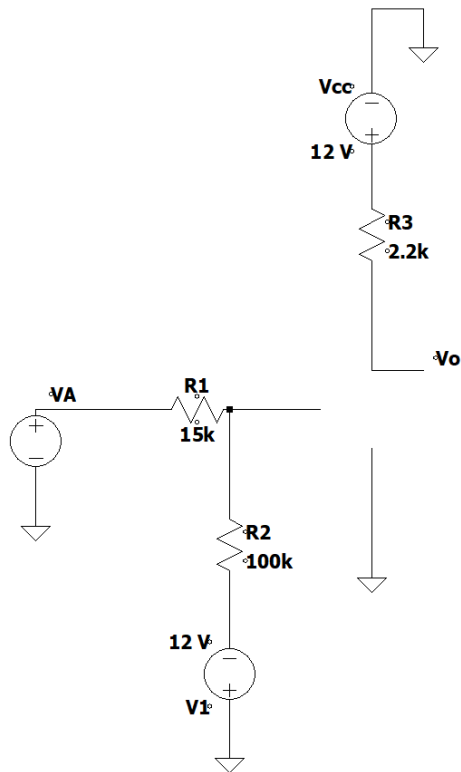
Exercise 2

Solution:

Step 1:

Case	BJT Q1
0	Cutoff
1	Saturation

Case 0:



Step 2:

$$I_B = I_C = I_E = 0$$

$$V_O = V_{CC} = 12\text{ V}$$

$$I_{R1} = I_{R2} = \frac{V_A - (-12)}{R_1 + R_2} = \frac{0.2 + 12}{15 + 100}\text{ mA}$$

$$= 0.106\text{ mA}$$

$$V_A = 0.2\text{ V}, V_1 = -12\text{ V}$$

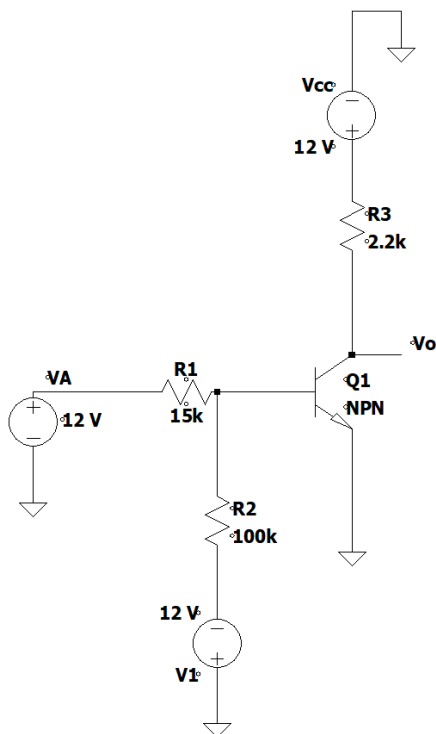
Current flows only through R_1 & R_2

Step 3:

$$P_0 = (V_A - V_1) \times I_{R1} = (0.2 - (-12)) \times 0.106\text{ mW}$$

$$= 1.2932\text{ mW}$$

Case 1:



Step 2:

$$V_O = V_{CE} = 0.2\text{ V}, V_B = V_{BE} = 0.8\text{ V}$$

$$I_{R1} = \frac{V_A - V_B}{15} = \frac{12 - 0.8}{15}\text{ mA} = 0.747\text{ mA}$$

$$I_{R2} = \frac{V_B - (-12)}{100} = \frac{0.8 + 12}{100}\text{ mA} = 0.128\text{ mA}$$

$$\text{Using KCL, } I_B = I_{R1} - I_{R2} = 0.619\text{ mA}$$

$$I_C = \frac{V_{CC} - V_O}{2.2} = \frac{12 - 0.2}{2.2}\text{ mA} = 5.363\text{ mA}$$

I_{R2} flows between V_A & -12 V

I_B flows between V_B & 0 V

I_C flows between V_{CC} & 0 V

Step 3:

$$\begin{aligned}
 \therefore P_1 &= (V_A - (-12)) \times I_{R1} + (V_B - 0) \times I_B + \\
 &\quad (V_{CC} - 0) \times I_C \\
 &= (12 + 12) \times 0.747 + (0.8 - 0) \times 0.619 + \\
 &\quad (12 - 0) \times 5.363 \\
 &= 82.7792 \text{ mW}
 \end{aligned}$$

$\therefore$ Maximum power dissipation = $P_1 = 82.7792 \text{ mW}$

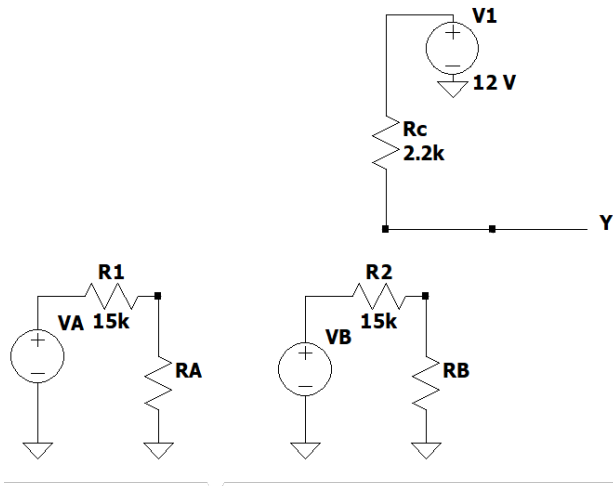
$\therefore$ Average power dissipation = $\frac{P_0 + P_1}{2} = 42.0362 \text{ mW}$

Exercise 3

Solution:

Both Q_1 & Q_2 will be off for (0,0).

Step 1:



Step 2:

$I_{R_C} = 0$ (open circuit)

$V_A = V_B = 0.2 \text{ V}$

Step 3:

$$P_{0,0} = 2 \times (V_A - 0) \times I_{R_A} = 2 \times 0.2 \times \frac{0.2 - 0}{15 + R_A}$$

$$\rightarrow 0.5 = 0.4 \times \frac{0.2}{15 + R_A}$$

$$\therefore R_A = R_B = 145 \text{ k}\Omega$$

Exercise 4

Solution:

Step 1:

$$V_{OH} = 11.5 \text{ V}, V_{OL} = 0.2 \text{ V}$$

Step 2:

Determination of V_{IH} :

V_{IH} is the lowest input voltage that would drive the RTL Not gate loads to forward active mode from saturation. Thus, the marginal condition here is,

$$\beta_{loads} = \beta_F = 30$$

For saturation mode, we know, $V_{BE} = 0.8 \text{ V}, V_{CE} = 0.2 \text{ V}$

$$\text{Thus, } I_{B_{loads}} = \frac{V_{IH} - V_B}{0.45} = \frac{V_{IH} - (0.8 - 0)}{0.45} = \frac{V_{IH} - 0.8}{0.45}$$

$$\text{And } I_{C_{loads}} = \frac{V_{CC} - V_{CE}}{0.64} = \frac{3.6 - 0.2}{0.64} = 5.3125 \text{ mA}$$

$$\text{Now, } \beta_{loads} = \frac{I_{C_{loads}}}{I_{B_{loads}}} = 30 \rightarrow \frac{V_{IH} - 0.8}{0.45} = \frac{5.3125}{30} \rightarrow V_{IH} = 0.88 \text{ V}$$

Determination of V_{IL} :

V_{IL} is the highest input voltage that would turn the RTL Not gates on & violate cutoff condition of $V_{BE} < 0.5 \text{ V}$ So, the marginal $V_{BE} = V_B = 0.5 \text{ V}$

Since $V_E = 0 \text{ V}$,

$$V_{BE} = V_B = V_{IL} - I_B \times 0.45 = V_{IL} - 0 \times 0.45 = V_{IL} (\text{In cutoff } I_B = 0)$$

$$\text{Thus, } V_{IL} = 0.5 \text{ V}$$

Step 3:

$$N_H = V_{OH} - V_{IH} = 3.5 - 0.88 = 2.62 \text{ V}$$

$$N_L = V_{IL} - V_{OL} = 0.5 - 0.2 = 0.3 \text{ V}$$

$$\therefore N_M = \min(N_H, N_L) = 0.3 \text{ V}$$

Exercise 5

Solution:

a) Step 1:

The current from the driver circuit is going outward to the NOT gate loads. Hence, the condition for maximum fanout is supply & demand current balance.

Step 2:

Case (1,1):

Both T_1 & T_2 on, output = V_{OL}

Supply current from the driver, $I_{RC} = \frac{V_{CC} - V_{OL}}{R_C} = \frac{3.6 - 0.2}{0.64} = 5.3125 \text{ mA}$

Loads will be cutoff due to low voltage at the base. Thus, demand current, $I_L = 0$.

$\therefore \text{Fanout} = \infty$

Case (0,1):

T_1 off & T_2 on, output = V_{OH}

Supply current from the driver, $I_{RC} = \frac{V_{CC} - V_{OH}}{R_C} = \frac{3.6 - 1.3}{0.64} = 3.6 \text{ mA}$

Loads will be in saturation due to high voltage at the base. Thus, demand

current, $I_L = \frac{V_{OH} - V_B}{R_B}$
 $= \frac{1.3 - 0.8}{0.45} = 1.11 \text{ mA}$

$\therefore \text{Fanout} = \text{floor} \left(\frac{I_{RC}}{I_L} \right) = 3$

Case (1,0): Same as Case (0,1)

Case (1,1): Same as Case (0,1)

Step 3: $\therefore \text{Maximum fanout} = \min(\infty, 3) = 3$

b) Applying KCL at output node,

$$I_{RC} = N \times I_L \rightarrow \frac{V_{CC} - V_o}{R_C} = 5 \times \frac{V_o - V_B}{R_B} \rightarrow \frac{3.6 - V_o}{0.64} = 5 \times \frac{V_o - 0.8}{0.45} \rightarrow V_o = 1.145 \text{ V}$$

c) For $(\beta_F)_{min}$, $V_{CE} = 0.2 \text{ V}$, $V_{BE} = 0.8 \text{ V}$

$$I_B = \frac{V_O - V_B}{R_B} = \frac{1.145 - 0.8}{0.45} = 0.767 \text{ mA}$$

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{3.6 - 0.2}{0.64} = 5.3125 \text{ mA}$$

$$\therefore (\beta_F)_{\min} = \frac{I_C}{I_B} = 6.93$$

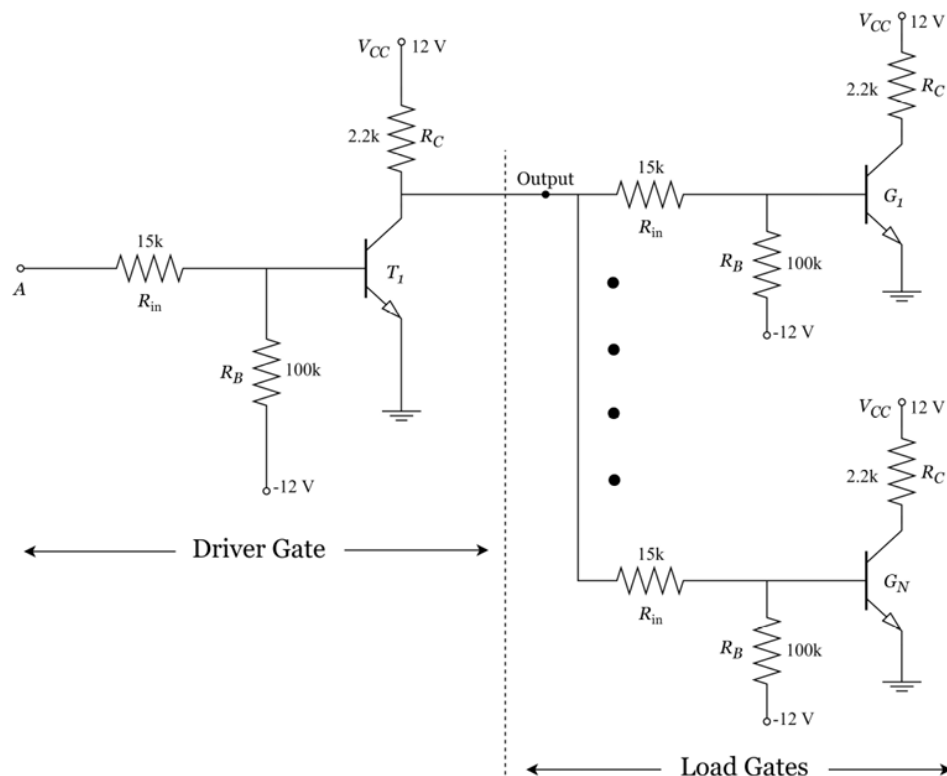
$$P_{loads} = 5 \times (V_{CC} - 0) \times I_C = 5 \times 3.6 \times 5.3125 = 95.625 \text{ mW}$$

$$\begin{aligned} \text{d) } P_{driver} &= \Delta V_{(base-ground)_A} \times I_{B_A} + \Delta V_{(base-ground)_B} \times I_{B_B} + \\ &\Delta V_{CC-ground} \times I_{R_C} \\ &= (V_A - 0) \times I_{B_A} + (V_B - 0) \times I_{B_B} + (V_{CC} - 0) \times I_{R_C} \\ &= (3.6 - 0) \times \frac{3.6 - 0.8}{0.45} + (3.6 - 0) \times \frac{3.6 - 0.8}{0.45} + (3.6 - 0) \times \frac{3.6 - V_{OL}}{R_C} \\ &= 2 \times 3.6 \times \frac{2.8}{0.45} + 3.6 \times \frac{3.6 - 0.2}{0.64} = 63.925 \text{ mW} \end{aligned}$$

$$\text{e) } Y = \overline{A + B}$$

Exercise 6

Solution:



a) Step 1:

The current from the driver circuit is going outward to the NOT gate loads. Hence, the condition for maximum fanout is supply & demand current balance.

Step 2:

Case (0):

T_1 off, loads on due to high voltage at base.

$$\text{Supply current from driver, } I_{RC} = \frac{12 - V_{OH}}{R_C} = \frac{12 - 10}{2.2} = 0.91 \text{ mA}$$

$$\text{Demand current, } I_L = \frac{V_{OH} - V_B}{R_{in}} = \frac{10 - 0.8}{15} = 0.61 \text{ mA}$$

$$\therefore \text{Fanout} = \text{floor} \left(\frac{0.91}{0.61} \right) = 1$$

Case (1):

T_1 on, loads off due to low voltage at base.

$$\text{Supply current from driver, } I_{RC} = \frac{12 - V_{OL}}{2.2} = 5.363 \text{ mA}$$

$$\text{Demand current, } I_L = \frac{V_{OL} - (-12)}{15 + 100} = 0.106 \text{ mA}$$

$$\therefore \text{Fanout} = \text{floor} \left(\frac{5.363}{0.106} \right) = 50$$

Step 3:

$$\therefore \text{Maximum fanout} = \min(50, 1) = 1$$

b) KCL at output of driver,

$$I_{RC} = \frac{12 - V_o}{R_C} = N \times I_L = 2 \times \frac{V_o - V_B}{R_{in}} \rightarrow \frac{12 - V_o}{2.2} = 2 \times \frac{(V_o - 0.8)}{15} \rightarrow V_o = 9.46 \text{ V}$$

c) If the input is high, $V_{CE} = 0.2 \text{ V}$. Thus, we have,

$$\begin{aligned} P_{driver} &= (V_{cc} - 0) \times I_{RC} + (12 - 0) \times I_B + (12 - (-12)) \times I_{RB} \\ &= (V_{cc} - 0) \times I_{RC} + (12 - 0) \times I_B + (12 - (-12)) \times I_{RB} \\ &= 12 \times \frac{12 - 0.2}{2.2} + 12 \times \left(\frac{12 - 0.8}{15} - \frac{12.8}{100} \right) + 12.8 \times \frac{12.8}{100} \\ &= 74.86 \text{ mW} \end{aligned}$$

$$\begin{aligned} \text{d) } P &= (V_{cc} - V_o) \times I_{RC} + 2 \times [(V_o - 0.8) \times I_{R_{in}} + (0.8 - (-12)) \times I_{RB} + \\ &\quad (12 - 0) \times I_{RC} + (0.8 - 0) \times I_B] + (0.2 - (-12)) \times I_{R_{in}} \end{aligned}$$

$$= (12 - 9.46) \times \frac{12-9.46}{2.2} + 2 \times [(9.46 - 0.8) \times \frac{9.46-0.8}{15} + 12.8 \times \frac{12.8}{100} + 12 \times \frac{12-0.2}{2.2} + 0.8 \times (\frac{9.46-0.8}{15} - \frac{12.8}{100})] + 12.2 \times \frac{0.2-(-12)}{15+100} = 146.95 \text{ mW}$$

e) **Step 1:**

$$V_{OH} = 10 \text{ V}, V_{OL} = 0.2 \text{ V}$$

Step 2:

Determination of V_{IH} :

V_{IH} is the lowest input voltage that would drive the RTL NOT gate loads to forward active mode from saturation. Thus, the marginal condition here is,

$$\beta_{loads} = \beta_F = 30$$

$$\text{Now, } I_{Cload} = \frac{V_{CC}-V_{OL}}{R_C} = \frac{12-0.2}{2.2} = 5.363 \text{ mA}$$

$$I_{Bload} = \frac{V_{IH}-0.8}{15} - \frac{0.8-(-12)}{100}$$

$$\therefore \beta_F = 30 \rightarrow \frac{5.363}{\frac{V_{IH}-0.8}{15} - \frac{12.8}{100}} = 30 \rightarrow V_{IH} = 5.4015 \text{ V}$$

Determination of V_{IL} :

V_{IL} is the highest input voltage that would turn the RTL NOT gate on & violate cutoff condition of $V_{BE} < 0.5 \text{ V}$. So, the marginal $V_{BE} = V_B = 0.5 \text{ V}$

KCL at base,

$$\frac{V_{IL}-V_B}{15} = \frac{V_B-0}{100} \rightarrow V_{IL} = \frac{V_B}{100} \times 15 + V_B = 0.575 \text{ V}$$

Step 3:

$$N_H = V_{OH} - V_{IH} = 4.5985 \text{ V}$$

$$N_L = V_{IL} - V_{OL} = 0.375 \text{ V}$$

$$N_M = \min(N_H, N_L) = 0.375 \text{ V}$$

Exercise 7

Solution:

$$\text{a) } Y = \overline{A + B}. C$$

b) Step 1:

Current from load flows toward driver circuit. Thus, the condition for maximum fanout is determined by driving driver BJTs into forward active mode.

Maximum current from load comes when $C=0$. Thus, we will ignore $C=1$ cases.

Step 2:

Case (0,0):

High voltage at the negative terminal of D_1 , thus it remains off.

$$\therefore \text{Fanout} = \infty$$

Case (0,1):

$$V_X = V_{OL} = 0.2 \text{ V}, D_1 \text{ on}$$

$$\text{Now, } I_{C_{Q_2}} = I_{C_{R_C}} + N \times I_{D_1} = \frac{12-0.2}{2.2} + N \times \frac{12-0.2-0.7}{100+0.25} = 5.363 + 1.11N$$

$$I_{B_{Q_2}} = \frac{12-0.8}{15} - \frac{0.8-0}{100} = 0.739 \text{ mA}$$

$$\text{Thus, } \beta_{max} = \frac{I_{C_{Q_2}}}{I_{B_{Q_2}}} = \beta_F = 25 \rightarrow 5.363 + 1.11N = 25 \times 0.739$$

$$\rightarrow N = \text{floor}(13.112) = 13$$

Case (1,0): Same as Case (0,1)

Case (1,1):

$$V_X = V_{OL} = 0.2 \text{ V}, D_1 \text{ on}$$

$$I_{C_{Q_1}} + I_{C_{Q_2}} = 2I_{C_{Q_1}} = I_{C_{R_C}} + N \times I_{D_1} = \frac{12-0.2}{2.2} + N \times \frac{12-0.2-0.7}{100+0.25} \\ = 5.363 + 0.11N$$

$$\therefore I_{C_{Q_1}} = \frac{1}{2}(5.363 + 0.11N)$$

$$I_{B_{Q_1}} = I_{B_{Q_2}} = \frac{12-0.8}{15} - \frac{0.8-0}{100} = 0.739 \text{ mA}$$

$$\text{Thus, } \beta_{max} = \frac{I_{C_{Q_2}}}{I_{B_{Q_2}}} = \beta_F = 25 \rightarrow \frac{1}{2}(5.363 + 0.11N) = 25 \times 0.739$$

$$\rightarrow N = \text{floor}(287.15) = 287$$

Step 3:

$$\therefore \text{Maximum fanout} = \min(\infty, 13, 287) = 13$$

- c) Maximum power will be dissipated in the case (1,1,0). This case turns on all the diodes & BJTs.

$$\begin{aligned} P_{max} &= P_{driver} + P_{load} = (12 - 0.2) \times I_{RC} + (12 - 0) \times I_{R_5} \\ &= 12 \times \left[\frac{12-0.2}{2.2} + \frac{12-V_Y}{100} \right] \end{aligned}$$

KCL at Y,

$$\frac{(12-V_Y)}{100} = \frac{V_Y-0.2-0.7}{.25} + \frac{V_Y-0-0.7}{.25} \rightarrow V_Y = 0.81 \text{ V}$$

$$\therefore P_{max} = 65.7 \text{ mW}$$

Exercise 8

Solution:

- a) Q_A off; Q_B & Q_C in saturation

$$V_Y = V_{CE} = 0.2 \text{ V}$$

$$\text{b) } I = \frac{V_{CC}-V_Y}{R_C} = \frac{20-0.2}{10} = 1.98 \text{ mA}$$

$$\text{c) } I_2 = \frac{I}{2} = 0.99 \text{ mA}$$

$$\text{d) } I_3 = I_2 = 0.99 \text{ mA}$$

Exercise 9

Solution:

$$\text{a) } v_0 = 0.2 \text{ V}$$

$$\text{b) } I_c = \frac{15-0.2}{2.2} = 6.7272 \text{ mA}$$

$$\text{c) } I_1 = \frac{15-0.8}{15} = 0.94667 \text{ mA}$$

$$\text{d) } I_2 = \frac{0.8-(-15)}{100} = 0.158 \text{ mA}$$

$$\text{e) } I_B = I_1 - I_2 = 0.78867 \text{ mA}$$

Exercise 10

Solution:

- a) V_{IL} is the maximum input voltage that would keep the RTL inverter loads from turning on. Here, the condition for turning on RTL loads, $V_{BE} < 0.5 V$
Thus, KCL at the base of RTL loads gives,

$$\frac{V_{IL}-0.5}{15} = \frac{0.5-(-5)}{100} \rightarrow V_{IL} = 1.325 V$$

- b) V_{IH} is the minimum input voltage that would keep the RTL loads in saturation and prevent transition into forward active mode. The condition for saturation mode operation is, $\beta_{forced} < \beta_F$

$$\text{Or, } I_B = \frac{I_C}{\beta_F} = \frac{5-0.2}{2.2} \times \frac{1}{30} = 0.072 mA$$

$$\rightarrow \frac{V_{IH}-0.8}{15} - \frac{0.8-(-5)}{100} = 0.072$$

$$\rightarrow V_{IH} = 2.761 V$$

- c) $N_M = \min(N_H, N_L) = \min(V_{OH} - V_{IH}, V_{IL} - V_{OL})$
 $= \min(4 - 2.761, 1.325 - 0.2) = 1.125 V$

- d) Case (0):

Driver cutoff, loads in saturation.

$$I_{supply} = \frac{V_{CC}-V_{OH}}{2.2} = \frac{5-4}{2.2} = 0.454 mA$$

$$I_{demand} = \frac{V_{OH}-0.8}{15} = \frac{4-0.8}{15} = 0.213 mA$$

$$\therefore Fanout = floor\left(\frac{I_{supply}}{I_{demand}}\right) = floor\left(\frac{0.454}{0.213}\right) = 2$$

Case (1):

Driver in saturation, loads cutoff.

$$I_{supply} = \frac{5-0.2}{2.2} = 2.181 mA$$

$$I_{demand} = \frac{V_{OL}-(-5)}{15+100} = \frac{0.2+5}{115} = 0.0452 mA$$

$$\therefore Fanout = floor\left(\frac{2.181}{0.0452}\right) = 48$$

$$\therefore Maximum Fanout = \min(2, 48) = 2$$

Exercise 11

Solution:

D_A on, D_B, D_C off. Q off.

$$a) I_L + i_2 = i_1 \rightarrow \frac{V_P - 0.7 - 0.1}{2} + \frac{V_P - (-12)}{R_1 + R_B} = \frac{12 - V_P}{R_2} \rightarrow \frac{V_P - 0.8}{2} + \frac{V_P + 12}{15 + 100} = \frac{12 - V_P}{15}$$

$$\therefore V_P = 1.90 \text{ V}$$

$$\therefore I_L = \frac{V_P - 0.8}{2} = 0.5521 \text{ mA}$$

$$b) i_1 = \frac{12 - V_P}{15} = 0.673 \text{ mA}$$

$$c) i_2 = \frac{V_P + 12}{115} = 0.121 \text{ mA}$$

$$d) I_B = 0 \text{ mA}$$

$$e) I_C = 0 \text{ mA}$$

$$f) P = (12 - 0.1) \times I_L + (12 - (-12)) \times i_2 = 9.474 \text{ mW}$$

Exercise 12

Solution:

All diodes off, Q in saturation.

$$a) I_L = 0 \text{ mA}$$

$$b) i_1 = \frac{12 - 0.8}{15 + 15} = 0.37334 \text{ mA}$$

$$c) i_2 = \frac{0.8 + 12}{100} = 0.128 \text{ mA}$$

$$d) I_B = i_1 - i_2 = 0.245334 \text{ mA}$$

$$e) I_C = \frac{12 - 0.2}{2.2} = 5.41 \text{ mA}$$

$$f) P = (12 - (-12)) \times i_2 + (12 - 0) \times I_B + (12 - 0) \times I_C = 70.936 \text{ mW}$$

$$g) \beta_{min} = \frac{I_C}{I_B} = 22.051$$